

Statistical Report on Fertility Data in Singapore

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Abstract

Singapore has experienced a long-term decline in fertility, creating challenges for population planning, labour supply and demographic sustainability. This report examines whether time series models trained on annual Singapore fertility data from 1960–2012 can accurately forecast Total Fertility Rate (TFR) and Total Live Births (TLB) over the 2013–2024 testing period. The analysis applies log transformations and differencing to address non-stationarity, then compares ARIMA and period-12 SARIMA models using residual diagnostics, information criteria and out-of-sample forecast accuracy. Period-12 SARIMA models were included because of evidence of 12-year Chinese zodiac-related birth timing effects in Singapore. The preferred model for both TFR and TLB was SARIMA(1,2,1)(1,0,0)[12], which passed residual autocorrelation checks and produced the lowest testing-period forecast errors among the viable models. The model forecasted TFR more accurately than TLB, likely because TFR is a smoother fertility rate, while TLB is an absolute count affected by population size, immigration, marriage patterns and external shocks. Overall, the results show that SARIMA models can capture the broad downward fertility trend, but univariate time series models have limitations when demographic structure changes or unusual shocks occur.

Introduction

Singapore has long experienced a long-term decline in fertility of its population. This has many knock-on effects to the economic and social structures of the country. Decreasing fertility can raise concerns about an ageing population, lack of labour supply and demographic sustainability. Therefore forecasting fertility is important for the government’s future understanding of policy and planning.

This report focuses on two annual time series, Total Fertility Rate and Total Live Births. These numbers were sourced from Singapore Department of Statistics and Immigration and Checkpoints Authority. These series will be henceforth referred to as TFR and TLB respectively. TFR is the number of children an average woman would have in her lifetime if she experienced current age specific fertility rates. TLB is the total number of live births experienced in Singapore over the time period, in our case a year. TFR reflects overall fertility more directly, whereas TLB is affected by population size, marital rates, immigration and other demographic factors.

This leads us to our research question of **Can time series models trained on data from 1960–2012 accurately forecast Singapore’s continuing fertility decline over 2013–2024?**

ARIMA and SARIMA models are used as they can model dependence in series’ using past values, shocks and differencing of the series to handle non-stationarity. SARIMA models with period 12 will also be tested in this paper as the Chinese Zodiac follows a 12 year cycle and it is previously shown evidence of affecting the number of births in each given year. This is shown particularly in Dragon birth years (Agarwal et al., 2017). This phenomenon is attributed to cultural and social factors, such as increased fertility rates among couples who believe that having a child in a Dragon year brings good fortune. This effect provides a substantively motivated reason for using SARIMA models with a period of 12, as they can capture such cyclical patterns in the data.

The analysis first examines the original TFR and TLB series, then applies log transformations and differencing to address non-stationarity. Viable models are then fit using the 1960-2012 training data and are tested the data from 2013-2024. The test results are in the form of residual diagnostics, information criteria and forecast accuracy.

Methods

The modelling workflow was designed to answer our forecasting needs. The TFR and TLB series were extracted from the dataset, reshaped into annual time series from 1960–2024 and split into a training period from 1960–2012 and a testing period from 2013–2024.

Time plots and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests of the two series were used to check for trends, changes in variance and the transformations required before modelling. Both series showed evidence of non-stationarity and since ARIMA models require the time series to be stationary, transformation of the series was necessary prior to modelling.

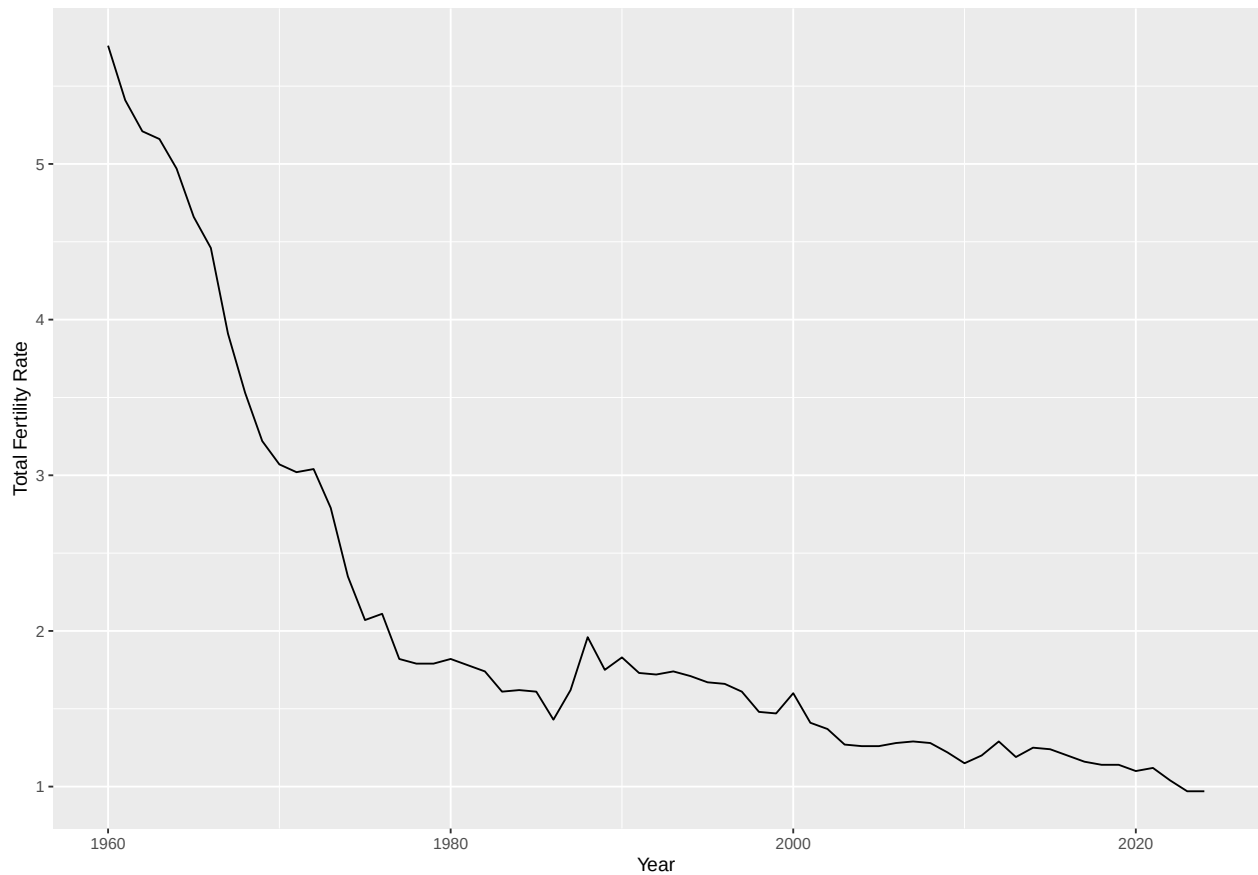


Figure 1: Annual Total Fertility Rate in Singapore



Figure 2: Total Live Births in Singapore

The log transformation was used to stabilise variance and make the model proportional rather than absolute changes. Differencing was used to remove the trends shown inside the data.

Both series were then transformed using either just the log or the log with first or second differencing. These transformed series were then tested by the KPSS test to ensure an improvement of the stationarity of the series.

Two transformations showed potential as being stationary (first-differenced and logged, and second-differenced logged). With the KPSS test a small p-value shows evidence against stationarity. For TFR, the KPSS test gave that $p = 0.0406$ for the first-differenced logged series and $p = 0.1$ for the second-differenced logged series. For the logged TLB with both first and second differencing gave the same p-values of $p = 0.1$. So for TLB stationarity was not rejected for either transformation.

Despite the first difference of the TFR data having a KPSS test result below the significance threshold, we would prefer there to be less differencing as we do not want to over-difference the data and introduce noise in a relatively short data-set. We kept the second-differenced models as comparison as well so we could see another set of models in the results.

Because the transformed series and baseline residual diagnostics showed dependence around lags 11–13, higher-lag ARIMA models were also considered. These models were included to capture longer-lag autoregressive dependence that was not represented by the low-order baseline models.

SARIMA-style models with period 12 were also considered. Although the data are annual, the period-12 structure was motivated by both the lag-12 dependence observed in the transformed series and the literature discussed in the Introduction on Chinese zodiac-related birth timing effects in Singapore. These models were included to assess whether a 12-year pattern could be represented more simply than by unrestricted

high-order ARIMA models. For the SARIMA candidates, the ACF and PACF plots of the transformed series were inspected to guide the initial period-12 model specification.

Models were only considered viable if their residuals were approximately white noise. This again was assessed with ACF and PACF, portmanteau tests and residual plots. Among viable models, the final model selection was based on AIC and forecast accuracy over the 2013-2024 period. Forecast accuracy was assessed with RMSE and MAE, with formulas provided in the appendix.

Results

The ACF and PACF plots of the log-first-differenced series are shown in Figure 3. For both TFR and TLB the plots show a remaining dependence on lags 11-13, particularly at lag 12. This suggests that low order ARIMA models may not fully account for the higher order dependencies shown by Figure 3.

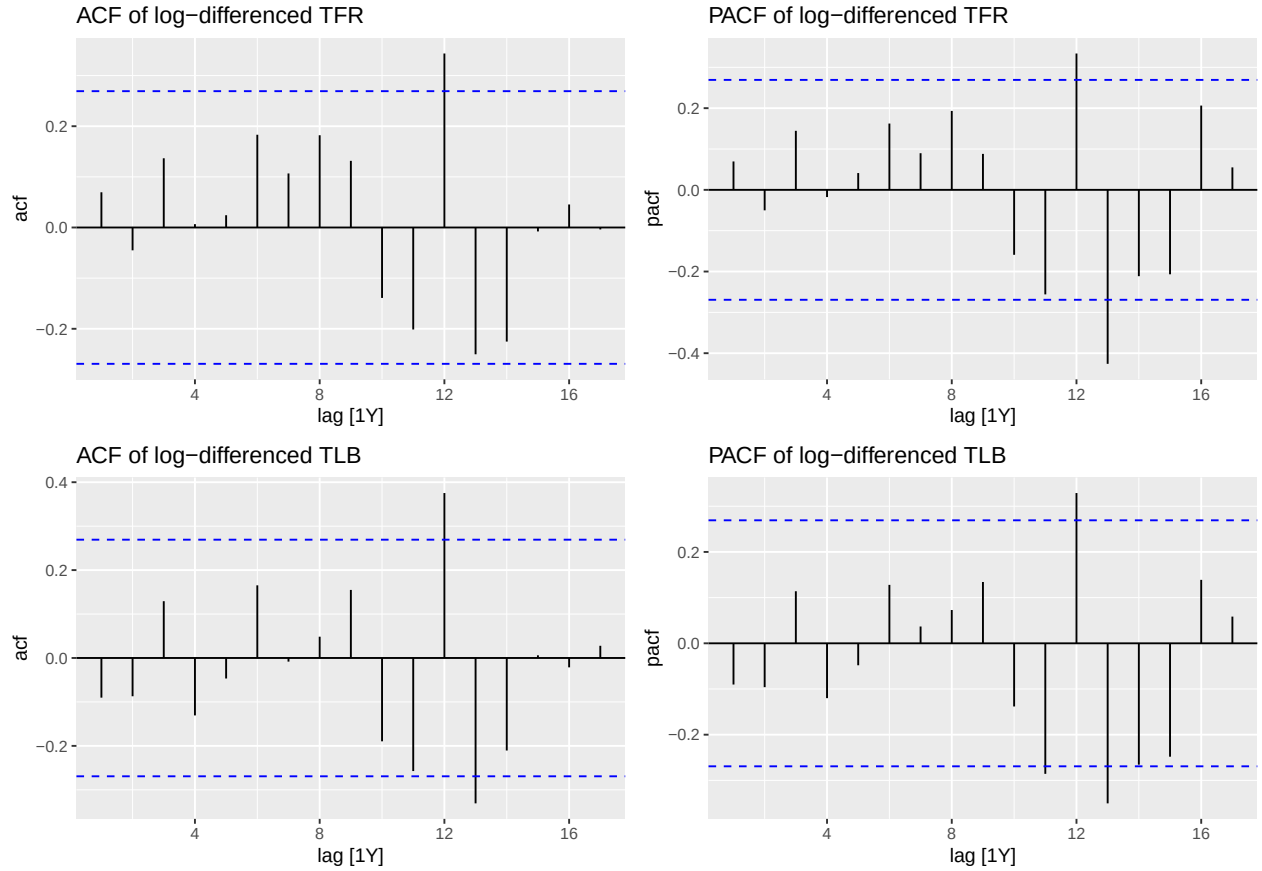


Figure 3: ACF and PACF plots of the log-first-differenced TFR and TLB series

To assess the period-12 SARIMA candidates, the ACF and PACF of $(1 - B)\log(X_t)$ (Figure 3) and $(1 - B^{12})(1 - B)\log(X_t)$ (Figure 4) were used to guide the structure of our models. For both series, Figure 3 showed PACF spikes around lag 12, this suggested a seasonal AR term could be useful, while the lower lag ACF structure supported testing simple MA terms. This informed our decision for our first SARIMA candidate model to be SARIMA(0,1,1)(1,0,0)[12]. Nearby candidates were also fitted by varying the non-seasonal AR and MA orders and by comparing seasonal AR and MA terms.

In our seasonally differenced plot (Figure 4), lag-12 dependence remained even after the period 12 differencing. This indicated that differencing alone could not remove the period 12 dependence structure and supported the use of seasonal ARIMA models.

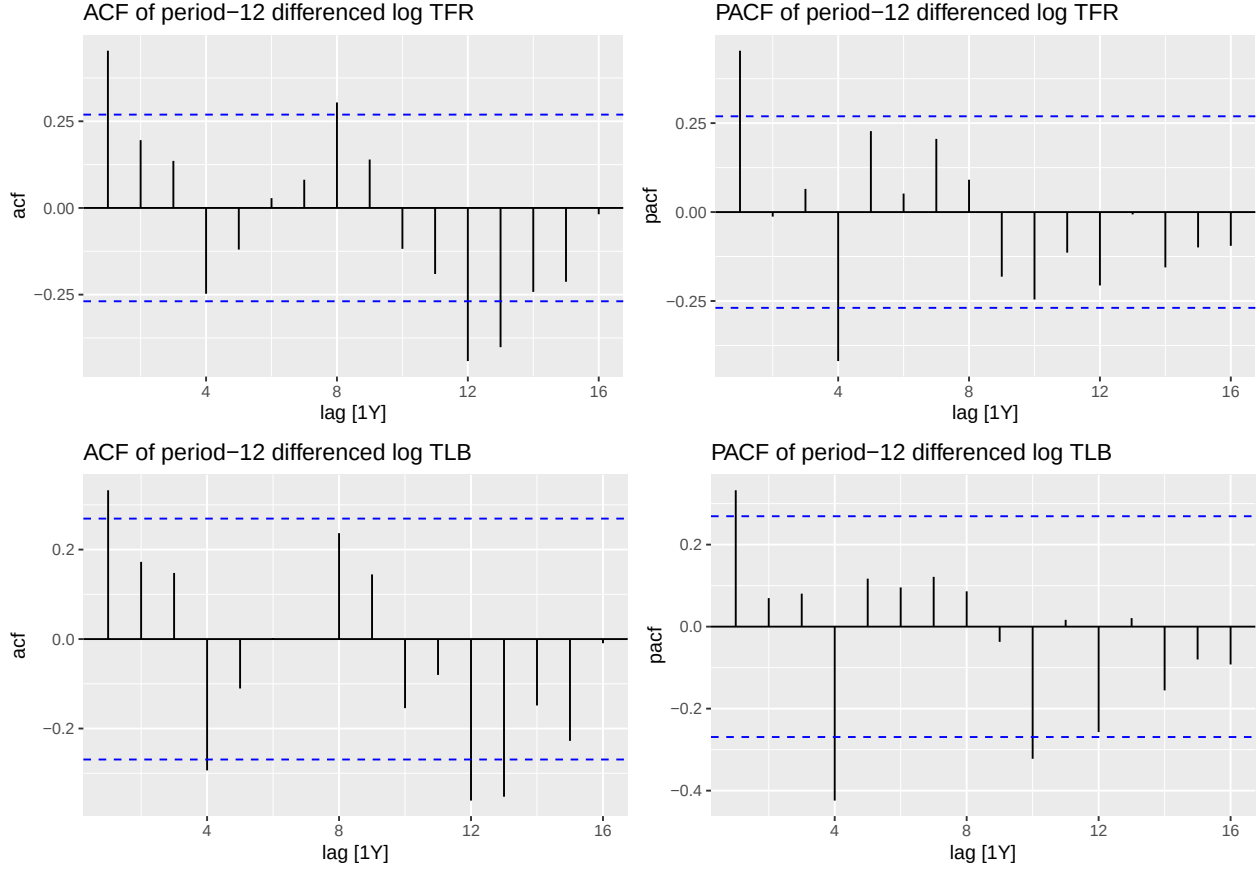


Figure 4: ACF and PACF plots of the log-first-differenced and period-12 differenced TFR and TLB series

Firstly, we fitted some low-order ARIMA models without seasonal terms, however, as expected based on the lag-12 dependence identified with the ACF/PACF plots these models failed to produce white noise residuals. We next decided to fit some higher-order ARIMA models with seasonal terms as well as some SARIMA models with lag 12.

Table 1: Ljung–Box residual tests for TFR and TLB candidate models

model_type	tfr_lb_pvalue	tfr_viable	tlb_lb_pvalue	tlb_viable
ARIMA(0,2,1)	0.005	No	0.003	No
ARIMA(1,2,1)	0.003	No	0.004	No
ARIMA(11,1,0)	0.000	No	0.001	No
ARIMA(12,1,0)	0.002	No	0.012	No
ARIMA(13,1,0)	0.020	No	0.045	No
ARIMA(13,1,1)	0.022	No	0.109	Yes
SARIMA(0,1,1)(0,0,1)[12]	0.027	No	0.028	No
SARIMA(0,1,1)(1,0,0)[12]	0.099	Yes	0.133	Yes
SARIMA(0,2,1)(1,0,0)[12]	0.004	No	0.049	No
SARIMA(1,1,1)(0,0,1)[12]	0.004	No	0.019	No
SARIMA(1,1,1)(1,0,0)[12]	0.108	Yes	0.115	Yes
SARIMA(1,2,1)(1,0,0)[12]	0.082	Yes	0.121	Yes

Table 1 reports the Ljung–Box portmanteau test p-values for the candidate models. Most high-order

ARIMA models with lags 11–13 failed the test for at least one series, with the only exception being ARIMA(13,1,1) for TLB. This suggests that adding unrestricted high-order AR terms was not sufficient to remove residual autocorrelation. The SARIMA models with a period-12 AR term performed better, with SARIMA(0,1,1)(1,0,0)[12], SARIMA(1,1,1)(1,0,0)[12] and SARIMA(1,2,1)(1,0,0)[12] being viable for both series.

Table 2: AIC, AICc and BIC values for TFR candidate models

.model	AIC	AICc	BIC
tfr_sar_011_100_log	-137.096	-136.596	-131.243
tfr_sar_111_100_log	-136.479	-135.628	-128.674
tfr_sar_121_100_log	-132.258	-131.388	-124.531

Table 3: AIC, AICc and BIC values for TLB candidate models

.model	AIC	AICc	BIC
tlb_sar_011_100_log	-146.736	-146.236	-140.882
tlb_sar_111_100_log	-145.079	-144.228	-137.274
tlb_sar_121_100_log	-138.646	-137.776	-130.918
tlb_ar13ma1_log	-141.446	-128.113	-112.178

Tables 2 and 3 show the AIC, AICc and BIC values for the viable TFR and TLB candidate models. For both series, SARIMA(0,1,1)(1,0,0)[12] had the lowest information criteria values among the viable candidates. However, these criteria should be interpreted with caution as the candidate sets both include models with different orders of differencing and as such cannot be directly compared across $d = 1$ or $d = 2$ models. With the research question in mind we are also preferring forecast accuracy for the 2013–2024 testing period as these in sample model fit results are not based on true out of sample forecast accuracy.

Table 4: Forecast accuracy for viable TFR models over 2013–2024

.model	RMSE	MAE	MAPE
tfr_sar_121_100_log	0.063	0.049	4.478
tfr_sar_011_100_log	0.098	0.073	6.945
tfr_sar_111_100_log	0.132	0.102	9.671

Tables 4 shows the forecast accuracy for the viable TFR models over 2013–2024 with 3 different forecasting metrics Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). In the results we can see that SARIMA(1,2,1)(1,0,0)[12] has the lowest for each of the testing criteria therefore being the best prediction model for TFR. This leads us to select it as our preferred TFR forecasting model.

Table 5: Forecast accuracy for viable TLB models over 2013–2024

.model	RMSE	MAE	MAPE
tlb_sar_121_100_log	3035.560	2280.238	6.151
tlb_sar_011_100_log	3667.483	2752.134	7.542
tlb_sar_111_100_log	3914.953	2943.136	8.091
tlb_ar13ma1_log	10016.593	8587.467	23.304

Table 5 shows the forecast accuracy for the viable TLB models over 2013–2024. Like in the TFR the trained model that performs best over the testing dataset is SARIMA(1,2,1)(1,0,0)[12]. It has the smallest of our 3 criterion and as such will be our final TLB prediction model as well.

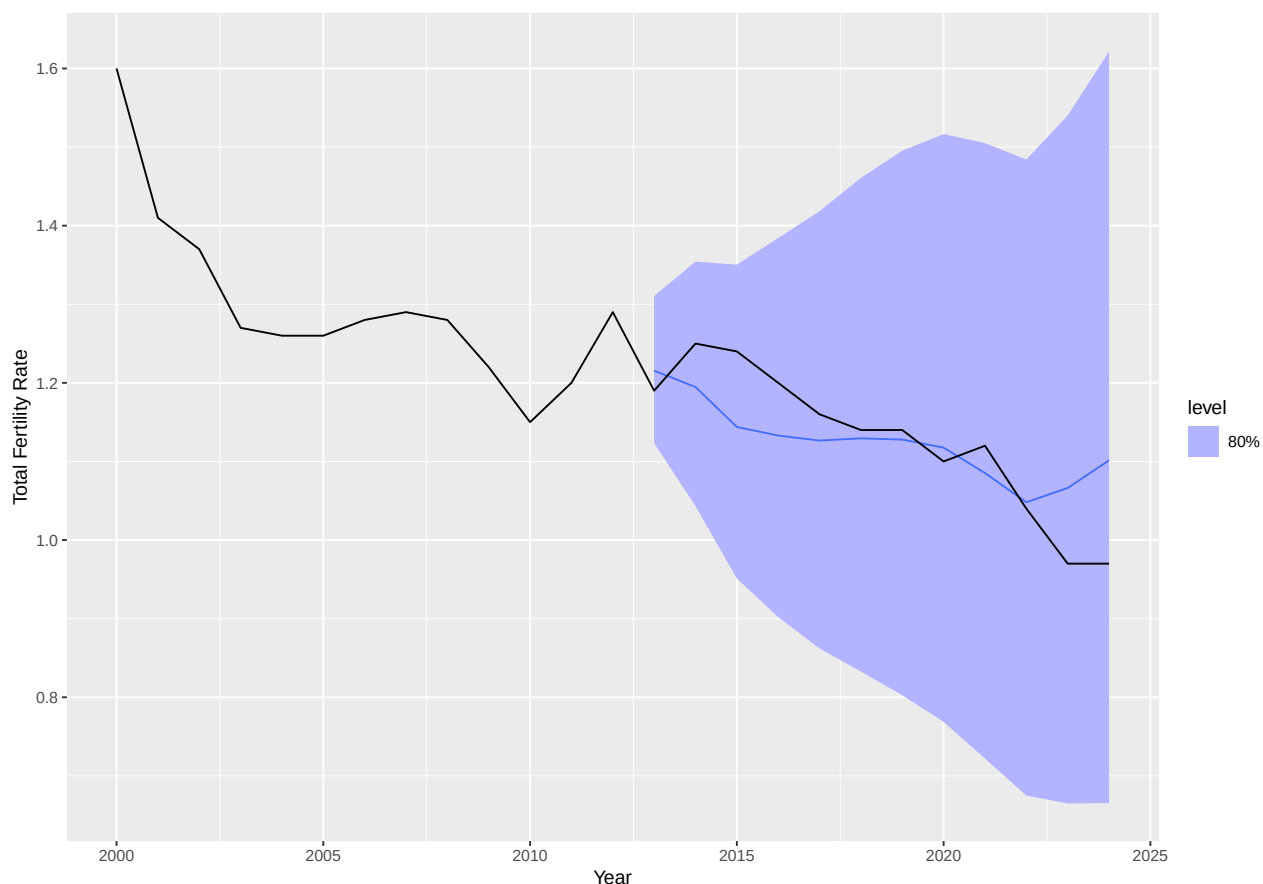


Figure 5: Selected TFR model forecast vs observed (2000–2024)

Figure 5 shows a plot of the chosen model for the TFR series, SARIMA(1,2,1)(1,0,0)[12], vs the actual observed data from 2000 to 2024. The black line is the observed data and the blue line is our predicted value for each year. The shaded blue area is our 80% prediction interval and shows the model forecast along with any uncertainty from the model. This figure supports the forecast accuracy results from Table 4. As we can see that our model captures the continual fall of the fertility rate during the testing period. However, the model does not perfectly match each year to year movement, and should be interpreted as being able to capture the general trend of the data downwards.

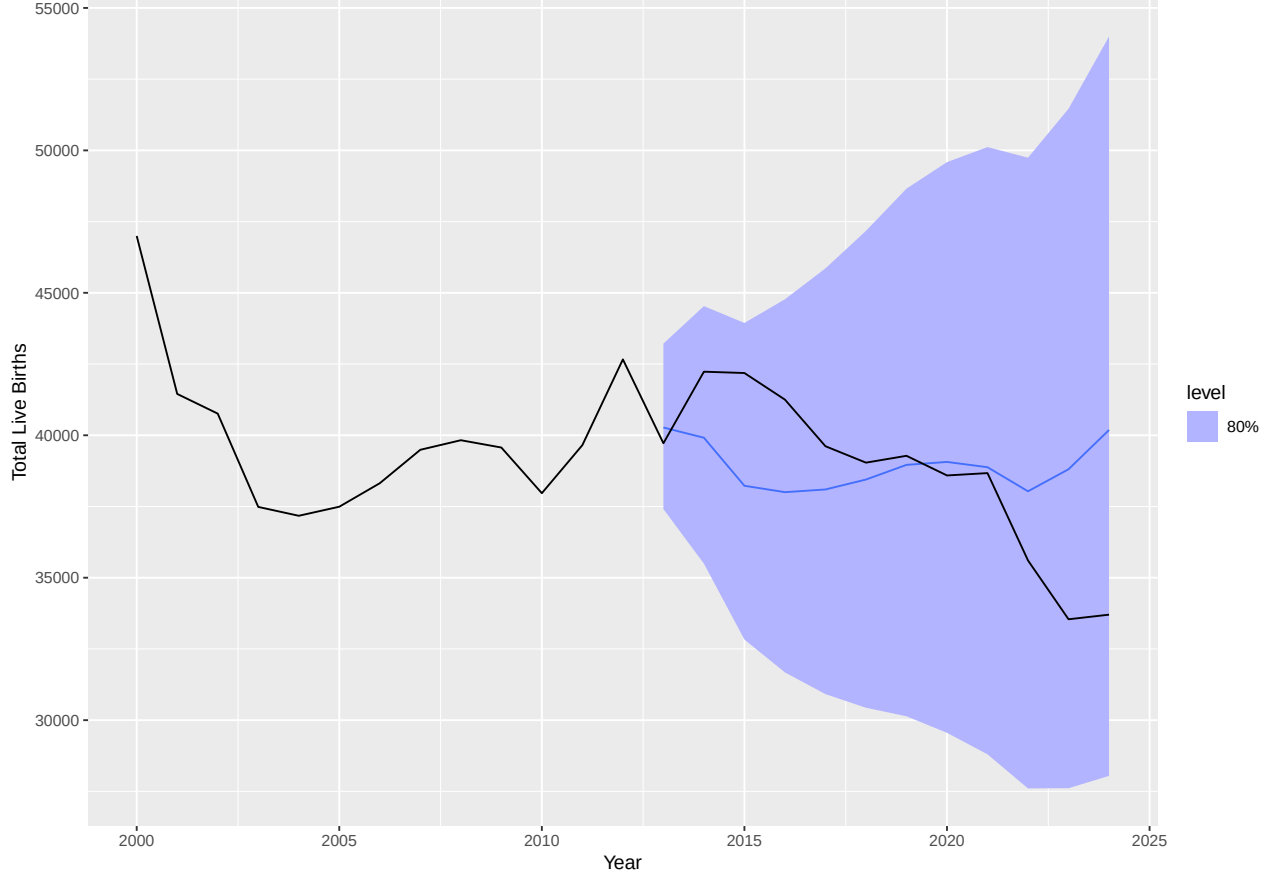


Figure 6: Selected TLB model forecast vs observed (2000–2024)

Figure 6 compares our selected model, $\text{SARIMA}(1,2,1)(1,0,0)[12]$, with the observed data from 2000 to 2024. The model forecasts a stabilisation in the TLB series, however, the observed series falls between 2021 and 2024. This shows that TLB is harder to forecast than TFR due to the chance that there may be more demographic variables which affect TLB that cannot be captured by a univariate time series model.

Overall, Figures 5 and 6 supports our final model choice of $\text{SARIMA}(1,2,1)(1,0,0)[12]$ for both series. The model passed the residual autocorrelation test and had the best testing sample forecast accuracy values out of our viable models.

Discussion

The main finding of this analysis is that a $\text{SARIMA}(1,2,1)(1,0,0)[12]$ is the preferred forecasting model for both TFR and TLB. The selection of a model with second differencing suggests that the decline of Singaporean birth rate and fertility is not just linear. Instead the decline appears to have steepened. This is evident in Figures 1 and 2, where the decline is especially pronounced in recent years. The $d = 2$ specification was able to react to the steepening decline better than the $d = 1$ was. Models which use $d = 1$ are generally more conservative and do not react as steeply. This could be an explanation for why our second-differenced model performed better for this short-term forecasting problem but we have to keep in mind that the model may be not as realistic over longer horizons. This is the general trade-off of using second-differencing: improved short-term responsiveness with the cost of stability in the long-term.

The model forecasts TFR more accurately than TLB. TLB being all the total live births in Singapore means that it is affected by total population and overall fluctuations in other demographic factors than just the birth rate. TFR is a more representative of the proportion of the population and is a rate which means that

it follows a smoother demographic trend rather than just population size. Therefore, a good model predicting overall fertility rates can still deviate when predicting total births as there are more variables.

We see this the sharp decline from 2021-2023 can be attributed to the overall shock from COVID-19 and the knock-on effect on immigration into Singapore. Singapore’s total population fell in 2020 and 2021 before rising again in 2022 (Singapore Department of Statistics, 2023). We must keep this in mind when analysing changes in TLB numbers. As spoken about earlier in the discussion the shock caused by COVID-19 may lead to a second-differenced model performing better as it is more flexible than single-differenced models.

The period-12 SARIMA component of the best performing models has captured the repeated 12-year dependence stemming from the Chinese Zodiac cycle. This is relevant in modelling as the previous evidence from about increased birth rates in specific Zodiac years, specifically high birth rates in Dragon years. We must keep in mind that our model is not directly modelling the cultural phenomena itself but is rather predicting changes in birth rates each year. The size of each Zodiac effect may vary across cycles and our simple seasonal model can only approximate this model.

The model also cannot account for the effect of pro-natalist policy interventions. This is relevant because Singapore has implemented multiple measures, including the Marriage and Parenthood Package, baby bonuses, tax measures, childcare subsidies and parental leave. Chen et al. (2018) argues that these policies may not fully target the groups most influential in determining TFR, particularly marriage and first-birth groups. This policy context is external to the univariate SARIMA models used in this report and future policies and other external factors will not be predicted by models like these.

In order to extend this analysis in birth rates of Singapore we could include multivariate forecasting models which could include other societal factors and may better predict changes in the overall birthrate. This would help separate the underlying fertility trend from temporary shocks and structural demographic changes.

Conclusion

This report found that SARIMA(1,2,1)(1,0,0)[12] was the preferred model for forecasting both Singapore TFR and TLB over the 2013–2024 testing period. The model had the best forecasting accuracy across all tested models and also produced white noise residuals. Overall, the results show that SARIMA models can capture the broad downward fertility trend, but univariate time series models have limitations when demographic structure changes or unusual shocks occur.

References

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Appendix

1. Time series notation and transformations

Let X_t denote an annual time series observed at year t . In this report, X_t represents either Total Fertility Rate (TFR) or Total Live Births (TLB). Since both series showed trend behaviour and changing variance, the log transformation was applied:

$$Y_t = \log(X_t).$$

The backshift operator B is defined as:

$$BY_t = Y_{t-1}.$$

First differencing was used to remove trend:

$$\nabla Y_t = Y_t - Y_{t-1} = (1 - B)Y_t.$$

Second differencing was also considered:

$$\nabla^2 Y_t = (1 - B)^2 Y_t.$$

To assess possible period-12 dependence, the following transformed series was also inspected:

$$(1 - B^{12})(1 - B)Y_t.$$

Although the data are annual, the period-12 structure was considered because of the 12-year Chinese zodiac cycle.

2. Autocorrelation and partial autocorrelation

For a stationary time series Y_t , the autocovariance function at lag h is:

$$\gamma(h) = \text{Cov}(Y_t, Y_{t-h}).$$

The autocorrelation function (ACF) is:

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}.$$

The ACF measures linear dependence between observations h time periods apart.

The partial autocorrelation function (PACF) measures the correlation between Y_t and Y_{t-h} after removing the linear effects of the intermediate lags $1, 2, \dots, h-1$. In this report, ACF and PACF plots were used to identify remaining dependence after transformation and to guide ARIMA and SARIMA model specification.

3. KPSS stationarity test

The Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test was used to assess stationarity. The null hypothesis of the KPSS test is that the series is stationary around a constant or deterministic trend. The alternative hypothesis is that the series is non-stationary.

The test is based on the decomposition:

$$Y_t = \mu_t + \epsilon_t,$$

where μ_t is a level or trend component and ϵ_t is a stationary error term.

The KPSS statistic is calculated using the partial sums of residuals:

$$S_t = \sum_{i=1}^t \hat{\epsilon}_i.$$

The test statistic is:

$$KPSS = \frac{1}{T^2 \hat{\sigma}^2} \sum_{t=1}^T S_t^2,$$

where T is the sample size and $\hat{\sigma}^2$ is an estimate of the long-run variance.

A small p-value provides evidence against stationarity. In this report, the KPSS test was used to compare the original, logged, log-first-differenced and log-second-differenced series.

4. ARIMA model

An ARIMA(p, d, q) model is written as:

$$\phi(B)(1 - B)^d Y_t = c + \theta(B)\varepsilon_t,$$

where:

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$$

is the autoregressive polynomial, and:

$$\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \dots + \theta_q B^q$$

is the moving average polynomial.

The error term ε_t is assumed to be white noise:

$$\varepsilon_t \sim WN(0, \sigma^2).$$

Here, p is the autoregressive order, d is the order of differencing, and q is the moving average order. Differencing is used to make the transformed series approximately stationary before fitting the ARMA structure.

5. SARIMA model

A SARIMA(p, d, q)(P, D, Q) $_s$ model extends ARIMA by including dependence at a specified period s . In this report, $s = 12$, corresponding to the 12-year Chinese zodiac cycle.

The general SARIMA model is:

$$\phi(B)\Phi(B^s)(1 - B)^d(1 - B^s)^D Y_t = c + \theta(B)\Theta(B^s)\varepsilon_t,$$

where:

$$\Phi(B^s) = 1 - \Phi_1 B^s - \Phi_2 B^{2s} - \dots - \Phi_P B^{Ps}$$

is the seasonal autoregressive polynomial, and:

$$\Theta(B^s) = 1 + \Theta_1 B^s + \Theta_2 B^{2s} + \dots + \Theta_Q B^{Qs}$$

is the seasonal moving average polynomial.

The selected final model was:

$$SARIMA(1, 2, 1)(1, 0, 0)_{12}.$$

For this model, $p = 1$, $d = 2$, $q = 1$, $P = 1$, $D = 0$, $Q = 0$, and $s = 12$. Therefore, the model can be written as:

$$(1 - \phi_1 B)(1 - \Phi_1 B^{12})(1 - B)^2 Y_t = c + (1 + \theta_1 B)\varepsilon_t.$$

This model was selected for both TFR and TLB because it passed residual autocorrelation checks and produced the lowest testing-period forecast errors among the viable candidate models.

6. Model estimation

The ARIMA and SARIMA models were estimated using maximum likelihood estimation. Under the assumption of normally distributed innovations,

$$\varepsilon_t \sim N(0, \sigma^2),$$

the likelihood function represents the probability of observing the data given the model parameters. The maximum likelihood estimates are the parameter values that maximise this likelihood, or equivalently maximise the log-likelihood:

$$\ell(\theta) = \log L(\theta),$$

where θ represents the model parameters, including AR coefficients, MA coefficients, seasonal coefficients and the innovation variance.

Models were compared using information criteria and testing-period forecast accuracy. The information criteria used were AIC, AICc and BIC.

The Akaike Information Criterion is:

$$AIC = -2\ell + 2k,$$

where ℓ is the maximised log-likelihood and k is the number of estimated parameters.

The corrected Akaike Information Criterion is:

$$AICc = AIC + \frac{2k(k+1)}{T-k-1},$$

where T is the sample size.

The Bayesian Information Criterion is:

$$BIC = -2\ell + k \log(T).$$

Lower AIC, AICc and BIC values indicate better in-sample fit after penalising model complexity. However, because the candidate models included different orders of differencing, these criteria were treated as supporting evidence rather than the sole basis for final model selection.

7. Ljung–Box portmanteau test

The Ljung–Box portmanteau test was used to test whether the residuals from a fitted model were approximately white noise. The null hypothesis is that the residual autocorrelations up to lag m are jointly zero:

$$H_0 : \rho_1 = \rho_2 = \cdots = \rho_m = 0.$$

The alternative hypothesis is that at least one residual autocorrelation is non-zero.

The Ljung–Box test statistic is:

$$Q = T(T+2) \sum_{h=1}^m \frac{\hat{\rho}_h^2}{T-h},$$

where T is the sample size, m is the number of lags tested, and $\hat{\rho}_h$ is the sample autocorrelation of the residuals at lag h .

Under the null hypothesis, the statistic is approximately distributed as:

$$Q \sim \chi_{m-k}^2,$$

where k is the number of estimated AR and MA parameters. A small p-value indicates remaining residual autocorrelation, meaning the model has not fully captured the dependence structure in the data. In this report, models with Ljung–Box p-values greater than 0.05 were treated as viable.

8. Forecast accuracy measures

Forecast accuracy was assessed on the testing period from 2013–2024. Let X_t be the observed value and $\hat{X}_t$ be the forecast value. The forecast error is:

$$e_t = X_t - \hat{X}_t.$$

The Root Mean Squared Error (RMSE) is:

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n e_t^2}.$$

The Mean Absolute Error (MAE) is:

$$MAE = \frac{1}{n} \sum_{t=1}^n |e_t|.$$

The Mean Absolute Percentage Error (MAPE) is:

$$MAPE = \frac{100}{n} \sum_{t=1}^n \left| \frac{e_t}{X_t} \right|.$$

Lower values of RMSE, MAE and MAPE indicate better forecast accuracy. RMSE penalises larger errors more heavily, while MAE gives the average absolute size of the forecast errors. MAPE expresses the error as a percentage of the observed value, making it easier to compare relative forecast accuracy across series.

9. Residual diagnostic plots

Residual diagnostic plots were used to visually assess the selected TFR and TLB models. The residual time plots were used to check whether the residuals fluctuated around zero without a clear trend. The residual ACF plots were used to check for remaining autocorrelation. The histograms were used to assess the approximate distribution of the residuals.

For both selected models, the residuals fluctuate around zero and the residual ACF plots show no strong systematic autocorrelation pattern, supporting the Ljung–Box test results used in the main report.

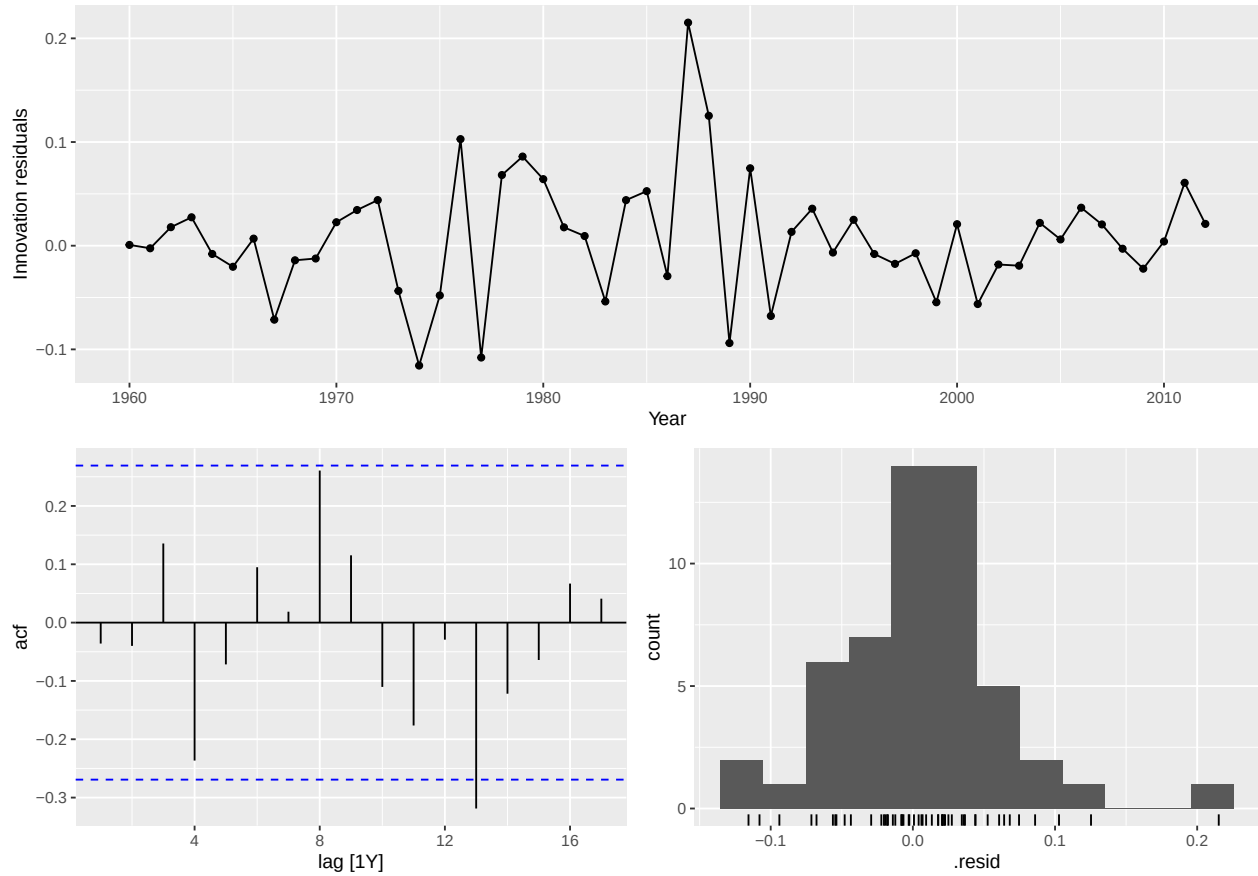


Figure 7: Residual diagnostics for selected TFR model

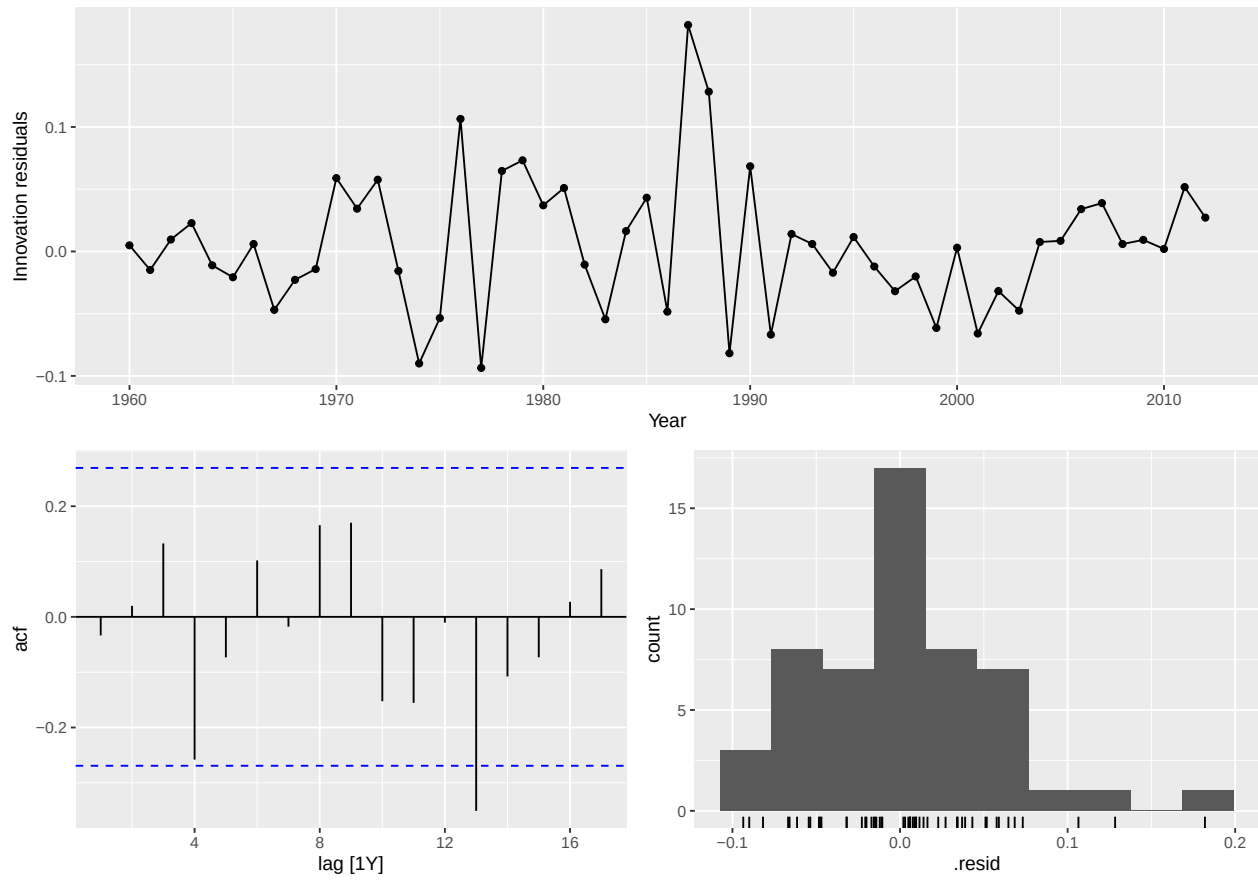


Figure 8: Residual diagnostics for selected TLB model